MEDROXYPROGESTERONE

3-monthly injection

Bone density loss

CHEMICAL NAME (6α)-17-(acetyloxy)-6-methylpregn-4-ene-3,20-dione (medroxyprogesterone acetate)

CLASS Progestogen (synthetic progesterone derivative)

TRADE NAMES Depo-Provera (IM) · Sayana Press (SC) · Provera (oral)

THERAPEUTIC USES / INDICATIONS

- Long-acting contraception (injectable every 12–13 weeks).
- Abnormal uterine bleeding, secondary amenorrhoea (oral).
- Endometriosis; endometrial protection in HRT.
- Hormone-sensitive cancers (high doses), hot flushes.

ADMINISTRATION

- **Depo-Provera:** deep IM (gluteal or deltoid) every 13 weeks; shake vial; do not massage site.
- **Sayana Press:** SC into abdomen/thigh every 13 weeks (self-injection possible).
- **PO** for gynaecological indications.

DOSAGES

- **Contraception:** 150 mg IM (or 104 mg SC) every 13 weeks; first dose days 1–5 of cycle.
- **Secondary amenorrhoea:** 10 mg daily for 5–10 days.
- **Endometrial protection:** 10 mg daily for 12–14 days per cycle.

PHARMACOKINETICS

- **Onset:** contraceptive immediately if given days 1–5. **Duration:** ≥3 months.
- Slow release from IM depot.
- Metabolised in liver. Return of fertility may take up to 1 year.

PHARMACODYNAMICS

- Inhibits **gonadotrophin secretion** → prevents ovulation; thickens cervical mucus; thins endometrium.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Irregular bleeding or amenorrhoea.
- Weight gain.
- Headache, mood changes, acne, ↓ libido.

ADVERSE EFFECTS (SERIOUS)

- Reduced bone mineral density (reversible; caution <18 yr, use >2 yr).
- Delayed return of fertility.
- VTE (high oncology doses).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy; undiagnosed vaginal bleeding.
- Current breast cancer; severe liver disease.
- Osteoporosis risk factors (relative); history of stroke/IHD (relative).

KEY INTERACTIONS

- Enzyme inducers have little effect on depot efficacy (unlike oral methods).
- **Aminoglutethimide:** ↓ levels.

NURSING MANAGEMENT

- Confirm not pregnant; record date and next due date (13 weeks).
- Deep IM technique; do not rub.
- Discuss bleeding changes, weight, bone health; review every 2 years.
- Provide reminder for next injection.

PATIENT EDUCATION

- Return every 13 weeks (up to 14 weeks still acceptable).
- Periods often become irregular or stop; this is expected.
- Fertility may take up to a year to return.
- Weight-bearing exercise and calcium help bones.

KEY POINTS

- Progestogen depot every 13 weeks.
- Irregular bleeding, weight gain, bone loss.
- 150 mg IM / 104 mg SC.
- Delayed return of fertility.

SOURCES CHECKED [679] Depo-SubQ Provera, SC (US label) [680] Medroxyprogesterone Acetate, oral (US label)

Full references p. 432–457